Total No. of Questions : 8] [Total No. of Printed Pages : 3

Paper Code : 21324

F- 424

B.C.A. (Fifth Sem.) Examination, 2025-26

(New Course)

Paper No - BCA-504

Software Engineering

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks. Symbols used have their usual meanings.

1. (a) What are "Software crises"? Discuss the evolution of software.

   (b) Explain the term "SDLC" specifying its role in software industry.

2. (a) What do you mean by spiral model of software development? Write down its pros and cons.

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   (b) Write down various ways of gathering requirements of customers. Give the detailed structure of SRS.

3. (a) Illustrate the features of a good software & explain each.

   (b) How cohesion is different from coupling? What are various types of cohesion.

4. (a) Describe different types of design approaches.

   (b) How modularity effects any design? Justify with one example.

5. (a) Explain "CASE Tools" in details.

   (b) What are various phases of system investigation?

6. (a) Elaborate "PERT" i.e. Program evaluation review technique in context with software development considering any relevant example.

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(b) How "Black-Box" testing is different from "White-box" Testing? Explain their types.

7. (a) Differentiate software engineering and software re-engineering.

(b) Discuss various types of maintenance. Why this phase is assumed to be the longest phase in SDLC models? Justify your answer.

8. Write in brief about any **four** of the following:

(a) Acceptance Testing

(b) Feasibility study

(c) Prototyping 4GL tools

(d) Test cases

(e) COCOMO method

(f) Project Management

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